

# Common $L^1$ basin boundaries for a bistable mean-field interval map

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## Abstract

We prove the common-basin-boundary conjecture stated by Keller in the 2009 Oberwolfach mini-workshop report on the spectrum of transfer operators. For the prescribed two-branch Möbius interval maps with feedback  $G(m) = A \tanh(Bm/A)$ ,  $0 < A \leq 2/5$  and  $6 < B \leq 16$ , the basin of the constant density equals the boundary of each noncentral stable basin in the relative  $L^1$  topology of all probability densities. We prove the stronger intermediate statement that the central basin is the  $L^1$  closure of the finite preimages of the constant density. A terminal cumulative transport is lifted backward with self-consistent feedback. A probability-space inverse contraction and uniform distortion bound make this construction valid for arbitrary integrable densities, including unbounded densities and densities with zeros. The global three-limit theorem and boundary seed are prior results of Bardet, Keller and Zweimüller.

## 1 The original question

Let  $I = [-1/2, 1/2]$ , with Lebesgue measure  $dx$ , and let

$$\mathcal{D} = \{u \in L^1(I) : u \geq 0 \text{ a.e., } \int_I u \, dx = 1\}, \quad \phi(u) = \int_I xu(x) \, dx.$$

All closures and boundaries below are relative to  $\mathcal{D}$  with its  $L^1$  metric. Fix  $0 < A \leq 2/5$  and  $6 < B \leq 16$ . For  $|r| \leq 2/5$  define

$$f_r(x) = \frac{(r+4)x + r + 1}{2rx + 2}, \quad T_r(x) = \begin{cases} f_r(x), & x < -r/4, \\ f_r(x) - 1, & x > -r/4. \end{cases}$$

The value at the cut is immaterial for densities. Write  $P_r$  for the density transfer operator of  $T_r$ , so  $(T_r)_*(u \, dx) = (P_r u) \, dx$ , and set

$$G(m) = A \tanh(Bm/A), \quad F(u) = P_{G(\phi(u))} u.$$

The question in [2, pp. 2713–2715, especially p. 2715] concerns exactly this parameter range and the full space  $\mathcal{D}$ .

We use the following established facts from [1, Theorem 2, Propositions 3 and 4]; they are also stated in [2, Theorem 2 and the discussion following it]. There are three fixed densities  $\mathbf{1}, u_+, u_-$ , every orbit in  $\mathcal{D}$  converges in  $L^1$  to one of them, and the basins of  $u_+$  and  $u_-$  are open. Moreover,  $\mathbf{1}$  is in both basin boundaries. The last fact follows from the earlier boundary theorem on the analytic integral-representation class. Put

$$W = \{u : F^n u \longrightarrow \mathbf{1} \text{ in } L^1\}, \quad B_{\pm} = \{u : F^n u \longrightarrow u_{\pm} \text{ in } L^1\}.$$

The cited theorem supplies  $\mathcal{D} = W \sqcup B_+ \sqcup B_-$ . In particular  $W$  is closed. The earlier analytic-core boundary theorem does not itself give a boundary theorem on all  $\mathcal{D}$ : that core is compact and is not  $L^1$  dense.

**Theorem 1.** *For the model and parameters above,*

$$W = \overline{\bigcup_{n \geq 0} F^{-n}(\{\mathbf{1}\})}^{\mathcal{D}} = \partial_{\mathcal{D}} B_+ = \partial_{\mathcal{D}} B_-.$$

*No boundedness, bounded variation, or positive lower bound on the densities is assumed.*

The new step is the full-density finite-preimage closure. The inverse expansion mechanism and invariant integral representations originate in [1, Sections 3–4]; we extend the inverse construction to an arbitrary probability space and give a simple exact continuous estimate. Sections 2–5 prove the theorem. No differentiable stable-manifold theorem on  $L^1$  is used.

## 2 An open map on all densities

For  $h_r(x) = (x + r/4)/(1 + rx)$ , direct algebra gives  $h_r^{-1} = h_{-r}$  and  $T_r = T_0 \circ h_r$  away from the cut. Let  $Q_r$  be pushforward of densities by  $h_r$ . It is a signed- $L^1$  isometry, with inverse  $Q_{-r}$ . The map  $(r, u) \mapsto Q_r u$  is jointly strongly continuous: this follows first for continuous functions from the smooth Jacobians, then for  $L^1$  functions by approximation and the isometry.

**Lemma 2.**  *$F : \mathcal{D} \rightarrow \mathcal{D}$  is a continuous open surjection. For either stable basin,*

$$F^{-1}(\partial_{\mathcal{D}} B_{\pm}) = \partial_{\mathcal{D}} B_{\pm}.$$

*Proof.* Write  $F = P_0 K$  with  $K(u) = Q_{G(\phi(u))} u$ . For  $v \in \mathcal{D}$  put  $M_v(r) = \int_I h_{-r}(x) v(x) dx$ . This decreases in  $r$ , since

$$\partial_r h_{-r}(x) = \frac{x^2 - 1/4}{(1 - rx)^2} \leq 0.$$

Thus  $r - G(M_v(r))$  increases by at least the parameter increment and changes sign between  $-A$  and  $A$ . Its unique zero  $\rho(v)$  satisfies  $|\rho(v) - \rho(w)| \leq (B/2) \|v - w\|_1$ , since  $|h_{-r}| \leq 1/2$  and  $\|G'\|_{\infty} \leq B$ . Consequently  $K^{-1}(v) = Q_{-\rho(v)} v$  is continuous. Checking the mean and scalar equation in both directions proves this is the inverse of  $K$ ; hence  $K$  is a homeomorphism.

For a prescribed  $w \in \mathcal{D}$  let  $v = P_0 w$ . On fibers with  $v(T_0 x) > 0$  put  $q(x) = w(x)/v(T_0 x)$ , and put  $q(x) = 1$  on the other fibers. Since

$$(P_0 w)(y) = \frac{1}{2} (w(y/2 - 1/4) + w(y/2 + 1/4)),$$

both branch values vanish on almost every zero fiber. It follows that  $0 \leq q \leq 2$  and  $P_0 q = \mathbf{1}$ , including there. For  $z \in \mathcal{D}$  define  $R_w(z)(x) = q(x) z(T_0 x)$ . Then

$$P_0 R_w(z) = z, \quad R_w(v) = w, \quad \|R_w(z) - w\|_1 = \|z - v\|_1.$$

These positive sections show that  $P_0$  is open and onto, in the relative topology. This proves the first assertion for  $F$ .

For  $B = B_+$  or  $B_-$ , tail convergence gives  $F^{-1}B = B$ . If  $x \in \partial B$ , continuity puts  $F(x)$  in  $\overline{B}$ , while  $x \notin B$  implies  $F(x) \notin B$ . Thus  $F(x) \in \partial B$ . Conversely, if  $F(x) \in \partial B$ , every relative neighborhood  $U$  of  $x$  has open image  $F(U)$  containing  $F(x)$ , hence meets  $B$ . Therefore  $U$  meets  $F^{-1}B = B$ ; since  $x \notin B$ , it is a boundary point.  $\square$

### 3 A self-consistent inverse contraction

Put  $J = [-1/2, 3/2]$ . The inverse of the lifted branch is

$$b_r(z) = f_r^{-1}(z) = \frac{2z - r - 1}{r + 4 - 2rz}, \quad f'_r(x) = \frac{4 - r^2}{2(1 + rx)^2}, \quad \partial_r b_r(z) = -\frac{1 - 4b_r(z)^2}{4 - r^2}.$$

On any probability space, for a bounded  $J$ -valued random variable  $Z$ , solve

$$r = G(\mathbb{E}b_r(Z)), \quad V(Z) = b_r(Z). \quad (1)$$

The residual is increasing by at least the parameter increment, because  $b_r$  decreases in  $r$ , and has opposite signs at  $\pm A$ . Thus the root exists uniquely. In particular its scalar derivative is at least 1.

**Lemma 3.** *On every real probability-space  $L^2$ , for bounded  $J$ -valued  $Z, Z'$ ,*

$$\|V(Z) - V(Z')\|_2 \leq \frac{21}{25} \|Z - Z'\|_2.$$

*The constant is uniform over the stated  $A, B$  range.*

*Proof.* Interpolate  $Z_t = (1 - t)Z + tZ'$ , and set  $X_t = V(Z_t)$ , with root  $r_t$ . For each fixed  $A > 0$ , scalar implicit differentiation and differentiation under expectation are valid: all derivatives of  $b_r$  are bounded on the compact rectangle  $J \times [-2/5, 2/5]$ , the feedback is smooth, and the residual derivative is at least 1. Put  $q = 1 - 4X_t^2$ ,  $g = G'(\mathbb{E}X_t)$ ,  $\beta = g/(4 - r_t^2)$  and let  $D$  multiply by  $f'_{r_t}(X_t)$ . Differentiating (1) gives the identity

$$\dot{X}_t = \left( \text{id} - \frac{\beta q \mathbb{E}}{1 + \beta \mathbb{E}q} \right) D^{-1} \dot{Z}_t. \quad (2)$$

This differentiates a bounded path, not a map on an unrestricted Banach-space neighborhood.

We first show that the rank-one factor in (2) has squared norm at most  $C = 1 + \beta/4$ . Let  $m = \mathbb{E}q$ ,  $s = \mathbb{E}q^2$ ; then  $m^2 \leq s \leq m$ . When  $s > m^2$ , in the orthonormal basis  $\mathbf{1}, (q - m)/\sqrt{s - m^2}$  the factor has matrix

$$L = \begin{pmatrix} (1 + \beta m)^{-1} & 0 \\ -\beta \sqrt{s - m^2}/(1 + \beta m) & 1 \end{pmatrix}.$$

It is the identity on the orthogonal complement. The lower diagonal entry of  $C \text{id} - L^*L$  is  $\beta/4$ , and direct expansion gives

$$\det(C \text{id} - L^*L) = \frac{\beta^2 [(Cm - 1/4)^2 + C(m - s)]}{(1 + \beta m)^2} \geq 0.$$

For  $\beta > 0$  these imply positive semidefiniteness. For  $\beta = 0$  the factor is the identity; constant  $q$  gives a scalar contraction on constants and identity on their orthogonal complement. This proves the claimed bound.

Set  $a = |r_t| \leq 2/5$ . The self-consistent tanh relation yields

$$0 \leq g = B(1 - r_t^2/A^2) \leq 16 - 100a^2, \quad \|D^{-1}\| \leq \frac{2 + a}{2(2 - a)}.$$

Consequently the squared norm of the derivative in (2) is bounded by

$$U(a) = \frac{(2 + a)^2}{4(2 - a)^2} \left( 1 + \frac{16 - 100a^2}{4(4 - a^2)} \right) = \frac{(2 + a)(4 - 13a^2)}{2(2 - a)^3} < \frac{7}{10}.$$

The strict inequality on the entire closed interval follows from

$$7(2-a)^3 - 5(2+a)(4-13a^2) = 172(a-13/43)^2 + 12/43 + 58a^3 > 0.$$

Integrate (2) along the path and use  $7/10 < (21/25)^2$ .  $\square$

We also need elementary uniform branch sensitivities, valid throughout the same rectangle:

$$\begin{aligned} 0 < \partial_z b_r &\leq \frac{3}{4}, & |\partial_r b_r| &\leq \frac{25}{96}, \\ |\partial_x \log f'_r(x)| &\leq 1, & |\partial_r \log f'_r(x)| &\leq \frac{35}{24}. \end{aligned} \quad (3)$$

Indeed  $1+rx \geq 4/5$ ,  $f'_r \geq 4/3$ , and  $|\partial_r b_r| = (1-4b_r^2)/(4-r^2) \leq 25/96$ . The last two logarithmic derivatives are  $-2r/(1+rx)$  and  $-2r/(4-r^2) - 2x/(1+rx)$ , respectively.

## 4 Backward correction of a central orbit

**Proposition 4.** *Every  $u \in W$  is an  $L^1$  limit of densities  $v_n$  with  $F^n v_n = \mathbf{1}$ .*

*Proof.* Work on the original probability space  $(I, u(x)dx)$ ; write  $\mathbb{E}$  for its expectation. Let  $X_0(x) = x$ ,  $X_{j+1} = T_{r_j}(X_j)$ ,  $r_j = G(\mathbb{E}X_j)$  and  $u_j = F^j u$ , the density of  $X_j$ . Let  $a_j \in \{0, 1\}$  be its branch label, so  $f_{r_j}(X_j) = X_{j+1} + a_j$ . All  $X_j$  are bounded;  $u$  need not be in  $L^2$ . Fix  $n \geq 1$  and set  $\epsilon = \|u_n - \mathbf{1}\|_1$ . Define

$$H_n(y) = -\frac{1}{2} + \int_{-1/2}^y u_n(t) dt, \quad Y_n = H_n(X_n).$$

$H_n$  is absolutely continuous, nondecreasing, onto  $I$ , fixes both endpoints, and  $\|H_n - \text{id}\|_\infty \leq \epsilon$ . The cumulative transform sends the nonatomic law of  $X_n$  to uniform law, even if  $H_n$  has flat intervals.

Recursively for  $j = n-1, \dots, 0$  define

$$Y_j = V(Y_{j+1} + a_j), \quad \rho_j = G(\mathbb{E}Y_j). \quad (4)$$

The branch-event sublaw of  $Y_{j+1}$  is dominated by its total law. If the latter is absolutely continuous, each such sublaw is too, and its smooth inverse-branch pushforward is absolutely continuous. Starting from uniform  $Y_n$ , induction proves that every  $Y_j$  has a density. The parameter in (4) is exactly its feedback, and  $T_{\rho_j}(Y_j) = Y_{j+1}$  almost surely: away from endpoints,  $b_{\rho_j}(Y_{j+1} + a_j)$  belongs to the branch  $a_j$ . Thus the density  $v_n$  of  $Y_0$  satisfies  $F^n v_n = \mathbf{1}$ . No independence of branch labels and later variables is assumed.

Uniqueness in (1) also gives  $X_j = V(X_{j+1} + a_j)$ . The same labels cancel in the input difference, so Lemma 3, with  $\kappa = 21/25$ , gives

$$\|Y_j - X_j\|_2 \leq \kappa^{n-j} \epsilon, \quad \Delta_j := |\rho_j - r_j| \leq 16\kappa^{n-j} \epsilon, \quad \sum_{j < n} \Delta_j \leq 84\epsilon. \quad (5)$$

The terminal estimate holds because  $H_n$  displaces by at most  $\epsilon$ ; the feedback estimate uses Cauchy-Schwarz on a probability space.

Define deterministic maps backward on each old branch by

$$H_j(x) = b_{\rho_j}(H_{j+1}(T_{r_j}(x)) + a_j(x)). \quad (6)$$

Here  $a_j(x)$  is the label for  $T_{r_j}$  at a point in  $I$ , not a new random label; evaluated at  $X_j$  it is the label used in (4). Since  $H_{j+1}$  fixes the endpoints, the two limits at the old cut in (6) are both

$b_{\rho_j}(1/2) = -\rho_j/4$ . Each  $H_j$  is therefore continuous, nondecreasing and onto, fixes the endpoints, and  $Y_j = H_j(X_j)$ . It is absolutely continuous: on each branch the inner old map is smooth bi-Lipschitz, the later map is absolutely continuous, and the outer inverse branch is smooth Lipschitz. The finitely many branches glue continuously. This argument permits flats and does not assume that arbitrary compositions of absolutely continuous maps are absolutely continuous.

Put  $d_j = \|H_j - \text{id}\|_\infty$ . Equations (3) and (6) give

$$d_j \leq \frac{3}{4}d_{j+1} + \frac{25}{96}\Delta_j, \quad d_n \leq \epsilon.$$

Iterating and then summing, using (5), yields

$$d_0 \leq \frac{183}{8}\epsilon, \quad \sum_{j < n} d_j \leq 3\epsilon + \frac{25}{24} \sum_{j < n} \Delta_j \leq \frac{181}{2}\epsilon. \quad (7)$$

Every fixed finite old cylinder map is smooth bi-Lipschitz onto its image, so it pulls exceptional derivative-null sets back to null sets. The chain rule applied on these cylinders gives, for Lebesgue a.e.  $x$ ,

$$H'_0(x) = u_n(X_n(x))R_n(x), \quad R_n(x) = \prod_{j < n} \frac{f'_{r_j}(X_j(x))}{f'_{\rho_j}(H_j(X_j(x)))}.$$

This makes sense for merely integrable  $u_n$ ; it divides by no density. Equations (3), (5), and (7) show

$$|\log R_n| \leq \sum_{j < n} d_j + \frac{35}{24} \sum_{j < n} \Delta_j \leq 213\epsilon. \quad (8)$$

The integral required next is against Lebesgue measure, not against  $u dx$ . For the external sequence of old parameters let  $c_n = P_{r_{n-1}} \cdots P_{r_0} \mathbf{1}$ . Then  $c_n \leq 2$ . For completeness, the source integral-representation identities [1, Section 4.1] give

$$w_y(x) = \frac{1 - y^2/4}{(1 - xy)^2}, \quad |y| \leq 2/3, \quad P_r w_y = p_r(y)w_{\sigma_r(y)} + (1 - p_r(y))w_{\tau_r(y)},$$

where

$$p_r(y) = \frac{1}{2} - \frac{r+y}{4+ry}, \quad \sigma_r(y) = \frac{2(y+r)}{(r+1)y+r+4}, \quad \tau_r(y) = \frac{2(y+r)}{(r-1)y-r+4}.$$

Direct substitution in the two inverse branches proves the identity. For  $|r| \leq 2/5$ ,  $|y| \leq 2/3$  the weight is in  $[0, 1]$ , both images remain in  $[-2/3, 2/3]$ , and  $1/2 \leq w_y \leq 2$ . Since  $\mathbf{1} = w_0$ , induction gives  $c_n \leq 2$ . Transfer duality and monotone truncation therefore imply

$$\int_I |u_n(X_n(x)) - 1| dx = \int_I |u_n(y) - 1| c_n(y) dy \leq 2\epsilon.$$

Together with (8) this proves

$$\|H'_0 - \mathbf{1}\|_1 \leq 2\epsilon e^{213\epsilon} + e^{213\epsilon} - 1. \quad (9)$$

Finally, any absolutely continuous nondecreasing onto map  $H : I \rightarrow I$  satisfies  $H_*(H' dx) = dx$ , by applying the fundamental theorem of calculus to antiderivatives of continuous test functions. Flat intervals are allowed. Write  $\|\cdot\|_{\text{TV}}$  for the full variation norm of a finite signed measure, equal to the  $L^1$  norm for its density. Pushforward contracts this norm. For continuous  $g$ , comparison with  $g(H)H' dx$  gives

$$\|H_*(g dx) - g dx\|_{\text{TV}} \leq \omega_g(\|H - \text{id}\|_\infty) + \|g\|_\infty \|H' - \mathbf{1}\|_1, \quad (10)$$

where  $\omega_g$  is its modulus of continuity. Although  $H_*(g dx)$  may have atoms, (10) is a measure inequality and remains valid. The actual  $H_{0*}(u dx) = v_n dx$  is already absolutely continuous by (4). For any continuous approximation  $g$  to the fixed  $u$ , (7), (9), and (10) yield

$$\|v_n - u\|_1 \leq 2 \|u - g\|_1 + \omega_g(183\epsilon/8) + \|g\|_\infty (2\epsilon e^{213\epsilon} + e^{213\epsilon} - 1).$$

As  $u \in W$ ,  $\epsilon \rightarrow 0$ . First fix  $g$  and let  $n \rightarrow \infty$ , then let its approximation error tend to zero. This proves  $v_n \rightarrow u$  in  $L^1$ , without any attraction-rate hypothesis.  $\square$

## 5 Completion, scope and verification

Every finite preimage of  $\mathbf{1}$  belongs to  $W$ , and  $W$  is closed; together with Proposition 4 this proves the closure identity. The cited boundary seed and Lemma 2 place every such preimage in both basin boundaries. Boundary closedness then gives  $W \subseteq \partial_{\mathcal{D}} B_{\pm}$ . Conversely, no point of either disjoint open stable basin can be in either boundary, and global three-limit convergence exhausts  $\mathcal{D}$ . Thus  $\partial_{\mathcal{D}} B_{\pm} \subseteq W$ , completing Theorem 1.

The result concerns the specified feedback and rectangle; it does not resolve the separate finite-particle limiting-mixture question, extend to  $A = 0$  or  $B = 6$ , or assert a nonlinear  $L^1$  differentiable stable manifold. The proof is analytic. The accompanying verification supplement retains the submitted exact identities and interval certificate, the sharper polynomial certificate above, rough-density/zero-fiber controls, source identities, and a dated bounded priority audit. Numerical controls are labeled corroboration and are not used to prove the infinite-dimensional theorem. The source audit distinguishes accessible preprint versions from the final journal version, whose proof text was not obtained. The original OWR statement postdates the journal publication and explicitly formulates this conjecture.

**AI use and review status.** AI tools were used extensively for proof development, verification, literature research, manuscript preparation, and adversarial review. The independent adversarial reviews were performed by AI agents. This is an unrefereed preprint and has not undergone independent external human peer review. Responsibility for its contents rests with the author.

## References

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